

Projection of sea ice conditions in Nunatsiavut (Labrador)

Thomas Amo Kyeimiah¹, Bruno Tremblay¹, Adrienne Tivy², Jean-François
Lemieux³, Mathieu Plante³, Joey Angnatok⁴

¹Department of Atmospheric and Oceanic Sciences, McGill University, Montréal, QC, Canada

²Canadian Ice Service, Environment and Climate Change Canada

³Recherche en Prévision Numérique Environnementale, Environnement et Changement Climatique

Canada, Dorval, QC, Canada

⁴Contractor, Nunatsiavut Research Center

Key Points:

- The CESM1.3-HR model projects a stable maximum March sea ice extent until 2060, followed by a rapid transition to ice-free conditions.
- The March sea ice thickness and the length of the ice-covered season are expected to decline gradually from 1 meter and 175 days to 0.5 meters and 75 days before the sharp decline in maximum sea ice extent occurs.
- High-resolution climate models are necessary for accurately capturing the coupling between deep water formation in the Labrador Sea and the sea ice response to climate change.

Corresponding author: Thomas Amo Kyeimiah, thomas.kyeimiah@mail.mcgill.ca

Abstract

Of all sub-regions of the Canadian Arctic, the Labrador Sea and Baffin Bay have the largest negative trend in summer sea ice area, raising important questions regarding the length of the “usable-sea-ice season” for coastal communities of Nunatsiavut. In this work, we focus on projected changes in sea ice thickness and extent in the Labrador Sea and Baffin Bay, and more specifically on changes in sea ice season length around five coastal communities, namely Nain, Hopedale, Makkovik, Postville, and Rigolet using output diagnostics from the High-Resolution (0.1 °) Community Earth System Model version 1.3 (CESM1.3-HR) for the period 1850-2100. Given that the fjords along the Labrador Coast where the coastal communities are located are not resolved in the model, the sea ice thickness and usable-sea-ice season length is derived from surface air temperature data (resolved by the model) and a simple freezing degree day model, validated using in-situ, reanalysis and remote sensing data. Results for the Baffin Bay and Labrador Sea show no significant trend in maximum March sea ice extent in the Labrador Sea followed by a rapid transition to winter ice-free conditions around 2060 at the time when the Arctic Ocean becomes seasonally ice-free and thick multi-year ice is no longer advected southward through the Nares Strait and along the Labrador Coastline. This is in contrast with the lower resolution CESM2 Large Ensemble (CESM2-LE) showing a re-expansion of the maximum sea ice extent starting in the middle of the 21st century followed by a sudden collapse at the end of the century due to a restratification of the Labrador Sea and shutdown of deep convection. Also of interest is the gradual decline in the “usable-sea-ice” season starting around 2020 from 150 days and a significant increase and inter-annual variation in sea ice season length in the mid-21st century - a signal that is robust to model spatial resolution and presence or absence of deep convection in the Labrador Sea. These results show the importance of resolving small-scale processes for regional climate projection.

1 Introduction

The Arctic is projected to become seasonally ice-free before the middle of this century, with less than 1000 GtCO₂ of additional anthropogenic emissions beyond 2019 levels, according to most climate models and across all emissions scenarios (Notz & Community, 2020). In the eastern Canadian sub-Arctic, significant warming has already been observed, with increases in annual, winter, and summer mean temperatures of approx-

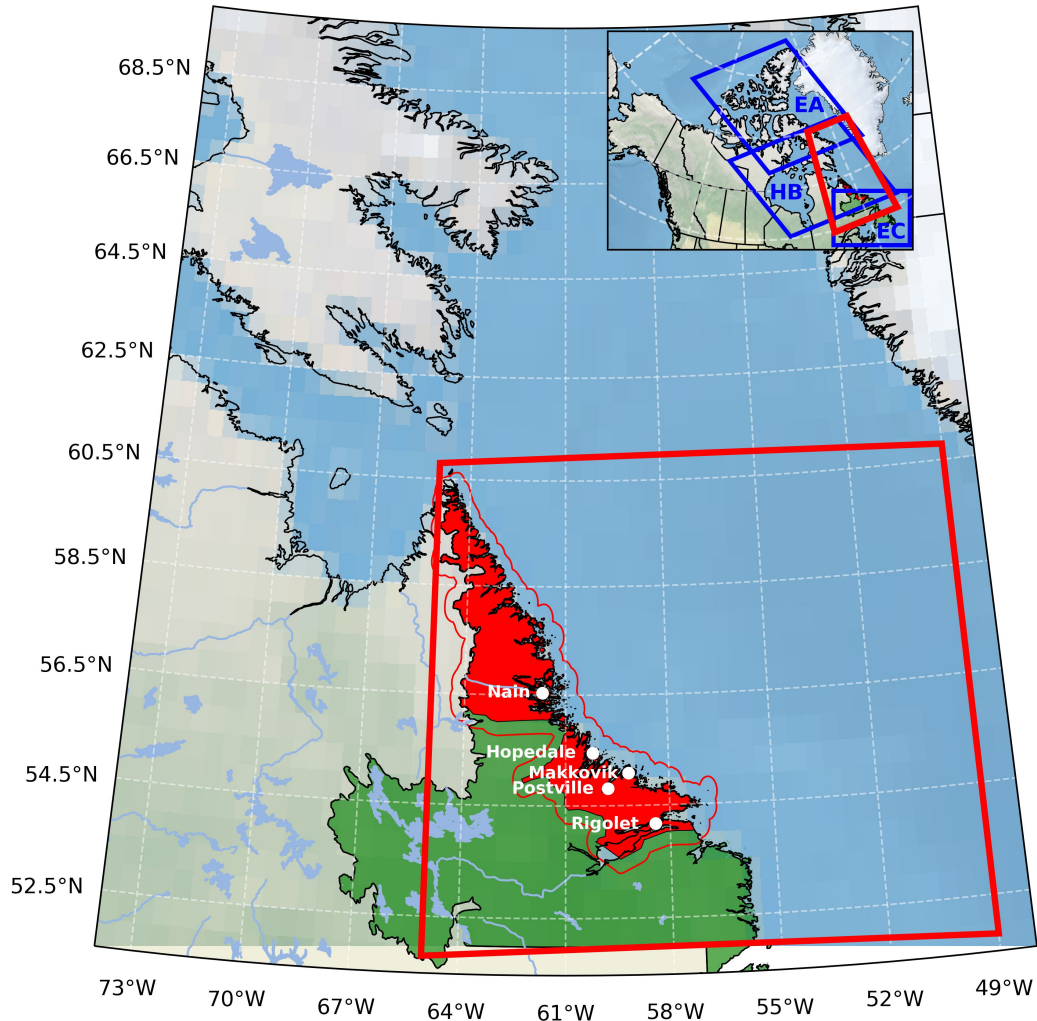


Figure 1. Map of Nunatsiavut (red), Labrador (green and red), and coastal communities (white dots) included in the ice season length analysis (see Figure 10). Inset: Canadian Ice Service regional ice chart areas covering the study region — Eastern Arctic (EA), Hudson Bay (HB), and East Coast (EC).

imately 1.5°C, 2.0°C, and 1.0°C, respectively, over the period 1881–2011 (Ouellet-Bernier et al., 2021). Among the six regions in the Canadian sub-Arctic that have experienced the most rapid sea ice decline between 1968 and 2010, two are located along the Canadian east coast: the Northern Labrador Sea (–15.7% per decade) and Baffin Bay (–11.6% per decade) (ECCC, 2023). These environmental changes are having tangible impacts on Arctic and sub-Arctic communities that depend on the natural environment for subsistence, travel, and recreation (Felt et al., 2012; ACIA, 2004; Nickels et al., 2005)

Nunatsiavut, meaning “our beautiful land” in Inuttitut, encompasses a large portion of the Labrador coast (Cunsolo et al., 2017). Rich in both cultural heritage and biodiversity, it is one of the Inuit regions where residents have observed environmental changes affecting their well-being (Felt et al., 2012; Cunsolo et al., 2017; CBC News, 2021; Newswire, 2021). For the Inuit of Nunatsiavut, sea ice shapes the rhythms of life and is central to their socio-ecological system. Viewed as an extension of the land, sea ice is vital for cultural practices, communication, mobility, and subsistence (Cuerrier et al., 2015; Ouellet-Bernier et al., 2021; Gearheard et al., 2006; Pearce et al., 2009). It functions not only as a symbol of cultural identity, but also as a conduit for community connection and a sensitive indicator of environmental change.

Sea ice in Baffin Bay and the Labrador Sea typically begins to form along the eastern Canadian sub-Arctic coast in November and retreats completely by August. The ice cover exhibits a west-to-east asymmetry with cold, fresher waters from the Canadian Arctic Archipelago flow southward along the Labrador coast, facilitating local sea ice formation and transporting multi-year ice from the Arctic via Nares Strait (Close et al., 2018; Kwok, 2005; Münchow, 2016; Howell et al., 2023). In contrast, relatively warmer Atlantic-origin waters move northward along the southwest coast of Greenland, maintaining largely ice-free conditions (Close et al., 2018; Wilbert Weijer and Thomas W.N. Haine and Ali H. Siddiqui and Wei Cheng and Milena Veneziani and Prajvala Kurtakoti, 2021). Of particular interest is the region of deep water formation in the Labrador Sea, where cold, salty waters convect to significant depths – a process highly sensitive to surface salinity conditions (Yashayaev et al., 2007; Yashayaev, 2024). For instance, results from a regional coupled ice–ocean model with specified atmospheric forcing project an increase in sea surface and bottom temperatures, a decrease in salinity, and a 70% reduction in winter sea ice extent over the Labrador and Newfoundland shelves for the period 2040–2069 compared to 1981–2010 (Han et al., 2019).

Several factors influence sea ice extent and surface salinity in the Labrador Sea, including synoptic and mesoscale storms, which are themselves modulated by the state of the North Atlantic Oscillation (NAO). These storms affect sea ice through enhanced offshore advection in regions with higher salinity (Häkkinen, 2002; Wu et al., 2023; Vaideanu et al., 2023). On longer timescales (interannual to decadal), the NAO also modulates sea ice conditions along the eastern Canadian coast by altering atmospheric circulation patterns –promoting anomalous northward advection of warm air over the northeast North

Atlantic and southward transport of colder air over Baffin Bay and the Labrador Sea (Deser et al., 2002, 2004; Eden & Willebrand, 2001). These sources of natural variability are superimposed on a greenhouse gas-driven warming trend, making it particularly challenging to disentangle internal climate variability from externally forced signals in this region near the NAO's center of action.

In this work, we examine projections of sea ice conditions in Nunatsiavut using the high-resolution Community Earth System Model (CESM1.3-HR), with a focus on the length of the ice season and the mechanisms driving its change. To assess the robustness of the high-resolution results, we also draw on simulations from a lower-resolution version of the same model, as well as in-situ and remotely sensed observations of sea ice thickness and concentration.

This paper is organized as follows. Section 2 describes the model diagnostics from the high- and low-resolution Community Earth System Models (CESM1.3-HR and CESM1.3-LR), the Large Ensemble of CESM2 (CESM2-LE), in-situ observations from Nain, and satellite data from the Canadian Ice Service. Section 3 presents the freezing degree-day model used to calculate sea ice extent and the length of the ice season for specific coastal communities at regional scales. Results and discussion are provided in Section 4. The main conclusions and directions for future work are summarized in Section 5.

2 Data

2.1 Sea ice concentration and thickness: Canadian Ice Service (CIS)

We use the Canadian Ice Service (CIS) regional ice chart database (CISDA; CIS Digital Archive) for the Eastern Arctic (EA), Hudson Bay (HB), and East Coast (EC) at monthly temporal resolution during winter, weekly resolution during summer prior to 2005; and weekly resolution all years thereafter (see Figure 1). CISDA is a collection of regional ice charts produced by ice analysts based on low and high resolutions satellite products (SSM/I sensors, MODIS, RADARSAT-1), aerial observations, ship reports, and operational models (Galley et al., 2012; Pizzolato et al., 2014; Tivy et al., 2011). The digital ice charts contain information on ice concentration, ice thickness, floe size, and ice type (e.g., landfast ice, first-year ice, multi-year ice, deformed ice, pack ice, Nilas ice) using the World Meteorological Organization's (WMO) Egg code. The data are stored using the ArcInfo Geographical Information System (GIS) following the Sea Ice GeoRef-

erenced Information and Data (SIGRID) standard guidelines; the raw data are gridded using nearest-point interpolation on a 10 x 10 km EASE grid from 1990 to December 2020 (JCOMM, 2004; Galley et al., 2012; Agnew & Howell, 2003). Note that the land masks change between different periods and the gridded data is stored on the most extensive landmask for consistency.

2.2 Sea ice concentration and thickness from CESM1.3-HR and CESM1.3-LR

We use the sea ice concentration, thickness, and 2 m temperature diagnostics from the International Laboratory for High-Resolution Earth System Prediction (iHESP), produced using both high-resolution and low-resolution configurations of the Community Earth System Model (Version 1.3, Meehl et al., 2019). iHESP is a collaborative effort involving Texas A&M University, the National Center for Atmospheric Research (NCAR), and the Qingdao Pilot National Laboratory for Marine Science and Technology (Chang et al., 2020). CESM1.3-HR includes four components: the Community Atmosphere Model version 5 (CAM5, Neale et al., 2010; Chang et al., 2020), the Parallel Ocean Program version 2 (POP2, Danabasoglu et al., 2012; Smith et al., 2010; Small et al., 2014; Chang et al., 2020), the Community Ice Code version 4 (CICE4, Hunke et al., 2010; Bailey et al., 2011; Small et al., 2014; Chang et al., 2020), and the Community Land Model version 4 (CLM4, Lawrence et al., 2011; Small et al., 2014; Chang et al., 2020).

The atmosphere and land components share the same grid with a nominal horizontal resolution of 0.25° ; the atmosphere model has 30 vertical levels with a model top at 3 hPa (Chang et al., 2020).

The ocean and sea-ice models are discretized using a tripolar B-grid with a nominal resolution of 0.1° which places two numerical poles over northern Canada and Russia to avoid the singularity at the geographic North Pole, while using a regular spherical polar grid in the Southern Hemisphere. POP2 has 62 vertical levels with a 10 m resolution in the first 155 m, gradually coarsening to 250 m near the maximum depth of 6000 m (Chang et al., 2020; Meehl et al., 2019; Zhang et al., 2020). The ocean model uses a 3rd order upwind advection scheme, a K-profile vertical mixing parameterization (Large et al., 1994), and explicitly resolves eddy-induced vertical transport. The sea ice component of the model includes the elastic-viscous plastic rheology (Hunke & Dukow-

icz, 2002), a subgrid-scale ice thickness distribution with multiple levels in each category (Bitz et al., 2001; Lipscomb, 2001), and an energy-conserving thermodynamic scheme that accounts for the effect of internal brine pockets (Bitz & Lipscomb, 1999).

The planned CESM1.3-HR project includes 10 ensemble members with one ensemble member covering the period from 1850 to 2100, and nine ensemble members covering the period from 1920 to 2100. The simulations were initialized from a 500-year pre-industrial control simulation. The model was forced with historical data for the period 1850 or 1920 until 2005 and with greenhouse gas from the Representative Concentration Pathway 8.5 from 2006-2100 (Chang et al., 2020). As of today, monthly diagnostic outputs are available for three of the ten ensemble members from 1920 to 2100. More details about CESM1.3-HR can be found in Chang et al. (2020).

2.3 Sea ice concentration: Community Earth System Model version 2 - Large Ensemble (CESM2-LE)

For context, we use the sea ice concentration from the lower resolution Community Earth System Model Large Ensemble version 2 (CESM2-LE). The CESM2-LE consists of 100 ensemble members at 1° spatial resolution covering the period 1850-2100 under CMIP6 historical and SSP370 future radiative forcing scenarios (Eyring et al., 2016; Rodgers et al., 2021). The CESM2-LE was initialized using a combination of macro and micro perturbations. Macro perturbations were implemented using 20 independent restart files at 10-year intervals from 1001–1191, from the preindustrial control simulation without explicit perturbations. The first ten members of the control simulation were chosen to minimize drift, while the last ten members include the extensive Mother Of All Runs (MOAR) output. Twenty micro-perturbations are applied to each of the years 1231, 1251, 1281, and 1301 of the preindustrial simulation, using small random perturbations $10^{-14} K$ applied to the atmospheric temperature field for a total of 80 micro-perturbation runs. The first half of the 100 members were subjected to the standard CMIP6 biomass-burning emissions. The other half used a temporally smoothed biomass burning emission inventory, achieved by implementing an 11-year running mean filter from 1990 to 2020 (Heyblom et al., 2022; Rodgers et al., 2021; Van Marle et al., 2017; CESM2-LE, 2023). Biomass burning emissions were temporally smoothed at each grid point following regridding onto the CESM2 grid. The goal was to preserve the total emissions as closely as possible while reducing the large interannual variability in the forcing (Rodgers et al., 2021).

2.4 Sea Ice Concentration: Climate Data Records

We use the monthly mean sea ice concentration (SIC) data from the National Snow and Ice Data Center (NSIDC) Climate Data Record (CDR, version 4). CDR-SIC is derived from the Scanning Multichannel Microwave Radiometer (SMMR), the Special Sensor Microwave/Imager (SSM/I), and the Special Sensor Microwave Imager/Sounder (SSMIS), flying on the Defense Meteorological Satellite Program (DMSP) satellites 5D-2/F11, 5D-2/F13, 5D-2/F8, 5D-3/F17, and the Nimbus-7 satellites. The data is stored on a 25 x 25 km polar stereographic grid centered on the North Pole from 25 October 1978 to today for the Arctic (Meier et al., 2021). The sea ice concentration products are based on the NASA Team and Bootstrap algorithms. The NASA Team algorithm compares brightness temperature (Tb) ratios from channels with horizontal (H) and vertical (V) polarization (19V, 19H, 37V) to identify surface types, using fixed tie-points and a weather filter (Cavalieri et al., 1984, 1997) while the Bootstrap algorithm uses Tb from 37H, 37V, and 19V channels, with dynamic tie points (Comiso, 1986; J. Comiso, 1995; Comiso & Nishio, 2008). Both algorithms have limitations, including reduced accuracy in regions with surface melt or thin ice, and in the marginal ice zone, where ice thermodynamic processes are key to changing sea ice conditions (Kern et al., 2020; Comiso, 2023; Agnew & Howell, 2003).

2.5 2 m Air Temperature: ERA5

We use 2-m air temperature data from ERA5, the fifth-generation global reanalysis produced by the European Centre for Medium-Range Weather Forecasts (ECMWF). ERA5 replaces the earlier ERA-Interim dataset and incorporates significant improvements, including higher temporal resolution (hourly data), enhanced model physics, and advanced data assimilation techniques (Hersbach et al., 2023; Olauson, 2018; Zhu et al., 2021). It combines a modern coupled model with a wide range of satellite and in situ observations to provide a comprehensive reconstruction of the atmosphere (Urban et al., 2021). ERA5 temperature data are provided on a regular 0.25° latitude-longitude grid for reanalysis and 0.5° for associated uncertainty estimates, with 137 hybrid pressure levels in the vertical. The 2-m temperature is derived by interpolating between the model's lowest level and the surface, accounting for near-surface atmospheric conditions (Hersbach et al., 2023).

2.6 Sea ice thickness: Nunatsiavut Research Center (Nain)

We use in-situ ice thickness and snow depth data collected by the Nunatsiavut Research Center in Nain. Measurements are taken monthly from early winter (January or February) through spring (May) at two sites—Taktok and Satosoak—located along the fjord connecting Voisey’s Bay to the Labrador Sea south of Nain. The dataset spans 2009 to 2023, with missing records for 2012 and 2014. Ice thickness is measured using a heated cable following the protocol of Mahoney et al. (2009), while snow depth is measured using snow stakes within a $10, \text{m} \times 10, \text{m}$ sampling area.

3 Methodology

3.1 Sea Ice Extent

Sea Ice Extent (SIE) is calculated as the total area of all grid cells where the sea ice concentration (SIC) exceeds 15%. Monthly mean SIE data from the Canadian Ice Service are computed as the average between the first and last days of each month prior to 2005, and as the mean of four weekly data points per month from 2005 onward

3.2 Mean Sea Ice Thickness and Volume

The weighted mean sea ice thickness in the Canadian Ice Service dataset (SIT_{CIS}) is estimated from the partial concentration and mean thickness of the thickest, second thickest, and third thickest ice stages, each of which encompasses multiple ice types with associated thickness ranges (Table 1). The weighted mean is computed as:

$$SIT_{\text{CIS}} = \frac{(CA \times SA) + (CB \times SB) + (CC \times SC)}{CA + CB + CC}, \quad (1)$$

where SA, SB, and SC are the mean thicknesses, and CA, CB, and CC the partial concentrations, of the thickest, second thickest, and third thickest ice stages, respectively (see Table 1).

The monthly Sea Ice Volume (SIV) in the Labrador Sea domain (Figure 1) is calculated as the product of the mean sea ice thickness for a given month and the grid cell area for all grid cells where the mean SIC is greater than or equal to 15%, as:

$$SIV_{\text{CIS}}(t) = \sum_{i,j} SIT_{i,j}(t) \times A_{i,j}(t), \quad (2)$$

Table 1. Stages of sea ice development, ice codes, thickness ranges, and mean ice thicknesses for each stage, as defined by the Canadian Ice Service (CIS). Values marked with an asterisk (*) are from the National Snow and Ice Data Center (NSIDC). (Glissenaar et al., 2021).

Development Stage	Ice Code	Description	Thickness (cm)	Mean Thickness (m)
3rd Thickest	99	No Data / Undetermined	N/A	0.13
	81	New Ice	< 10	
	82	Nilas, Ice Rind	< 10	
	83	Young Ice	10 - < 30	
	84	Grey Ice	10 - < 15	
	85	Grey-White Ice	15 - < 30	
2nd Thickest	86	First Year Ice	≥ 30	0.58
	87	Thin First Year Ice	30 - < 70	
	91	Medium First Year Ice	70 - < 120	
Thickest	93	Thick First Year Ice	≥ 120	2.68
	95	Old Ice	200 - 400*	
	96	Second Year Ice	200 - 400*	
	97	Multi-Year Ice	300 - 400*	

where $SIV(t)$ is the total sea ice volume at time t , $SIT_{i,j}(t)$ is mean ice thickness at grid point (i, j) and time t and $A_{i,j}(t)$ is the area of each grid cell (i, j) .

3.3 Length of ice season

We define the length of the ice season for a coastal community in Nunatsiavut as the number of days when sea ice is suitable for subsistence or recreational activities—specifically, from the day sea ice reaches a minimum thickness of 30 cm (start of season) to the onset of surface melt (end of season). The CESM1.3-HR however does not include the fjords along the coastlines. For this reason, we infer the sea ice thickness for different communities along the coast from the simulated (CESM1.3-HR) surface air temperature and a simple freezing degree day (FDD) model validated using ERA5 reanalysis surface air temperature data, date of first appearance of sea ice as per weekly Canadian Ice Service ice charts, and in-situ sea ice thickness measurements in Nain made by the Nunatsiavut Government (Nunatsiavut Research Center, 2024). The end of the sea ice season is simply set from the first thawing degree day in CESM1.3-HR.

To this end, we consider a simple 0D, steady state (linear temperature profiles) sea ice thermodynamic model. In this model, the temporal evolution of the sea ice thickness

256 can be written as:

$$\rho_i L_f \frac{dh_i}{dt} = F_c^i = F_c^s, \quad (3)$$

257 where ρ_i is the ice density ($= 917 \text{ kg/m}^3$), L_f is the latent heat of fusion ($= 3.35 \times 10^5 \text{ J/kg}$),
 258 h_i is the sea ice thickness, and F_c^i ($= F_c^s$) is the conductive heat flux in the ice.

259 The conductive heat flux through the snow-ice system in turn is equal (see Figure 2) to:

$$F_c^i = -K_i \frac{T_i - T_f}{h_i} = -K_s \frac{T_s - T_i}{h_s} = F_c^s, \quad (4)$$

260 where K_i and K_s are the thermal conductivity of ice ($= 2.0 \text{ W/m}\cdot\text{K}$) and snow ($= 0.2$
 261 $\text{W/m}\cdot\text{K}$), h_s is the snow depth, T_i , T_s and T_f are the snow-ice interface, snow surface
 262 and freezing point temperatures, and h_s is the snow depth. For simplicity, we assume
 263 that the surface (snow) temperature is equal to the 2 m surface air temperature, that
 264 the freezing temperature is constant at $T_F = 0^\circ\text{C}$, and that the snow depth is $h_s =$
 265 5, cm.

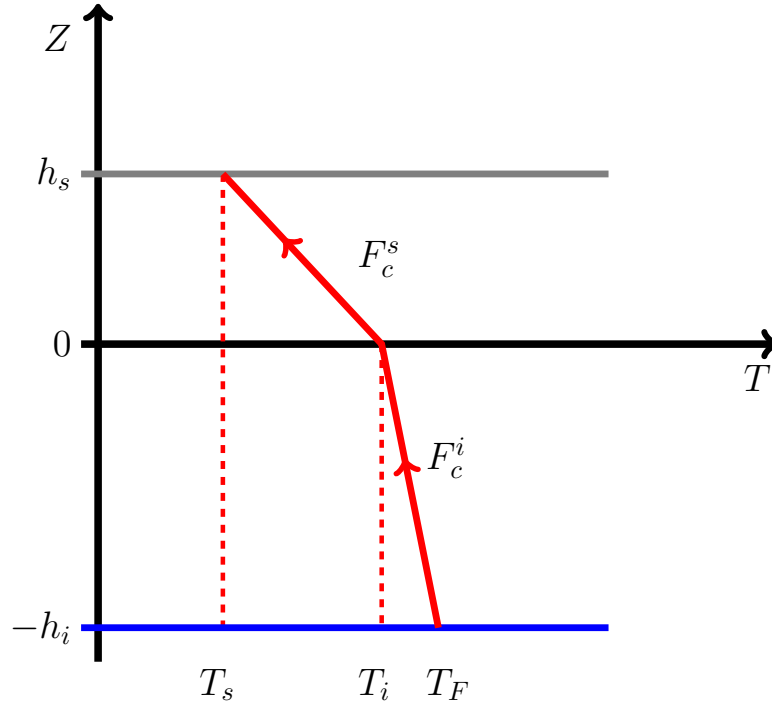


Figure 2. Schematic representation of the linear temperature profiles in a 0D steady-state sea ice thermodynamic model, showing the conductive heat flux through the ice (F_c^i) and snow (F_c^s) layers.

266 Solving for T_i from Equ. 4 and substituting F_c^i (or F_c^s) into Equ. 3, we obtain:

$$\rho_i L_f \frac{dh_i}{dt} = F_c = -\frac{K_i K_s}{K_i h_s + K_s h_i} T_s. \quad (5)$$

267 Direct integration from t ($= 0$, ice onset) to any t and from $h_i = 0$ m to $h_i(t)$ yields

$$h_i = \frac{-K_i h_s \pm \sqrt{(K_i h_s)^2 - 4K_s \frac{K_i K_s \text{FDD}}{\rho_i L_f}}}{2K_s}, \quad (6)$$

268 where FDD is the number of freezing degree days ($\int_0^t T_s dt$) converted from $^{\circ}\text{C} \cdot \text{day}$
 269 to $^{\circ}\text{C} \cdot \text{s}$.

270 4 Results

271 4.1 Historical

272 4.1.1 Seasonality

273 The simulated length of the sea ice season is in general agreement with the CIS ob-
 274 servation with the onset and disappearance of sea ice in November and August, respec-
 275 tively. Key differences include an earlier sea ice advance along the Labrador Coast in De-
 276 cember and a reduced maximum extent, particularly in the northern Labrador Sea (Fig-
 277 ure 3 and 4). The negative bias is in part due to deeper penetration of warm waters from
 278 the West Greenland Current (Figure 4). The overestimation of sea ice extent in early
 279 winter has been attributed by Chang et al. (2020) to excessive thermodynamic ice growth,
 280 along with the model's explicit parameterization of vertical ocean heat transport by mesoscale
 281 eddies, which reduces upward heat flux and delays oceanic heat release to the surface.
 282 The simulated change in maximum sea ice extent from the pre- to post-2000 period is
 283 also underestimated, with a reduction of only 0.02 million km^2 in the model compared
 284 to 0.06 million km^2 in the observations (Figure 3a). Specifically, CESM-HR maintains
 285 a relatively stable winter climate through the 20th and early 21st centuries and does not
 286 capture the pronounced observed decline in maximum sea ice extent in the late 20th cen-
 287 tury (Figure 5). Finally, the simulated interannual variability in maximum sea ice ex-
 288 tent is much smaller when compared with observations where large excursions from the
 289 mean are sometimes present (Figure 4 and 5). The simulated changes in sea ice volume
 290 are in general agreement with observations (considering the uncertainty in the CIS es-

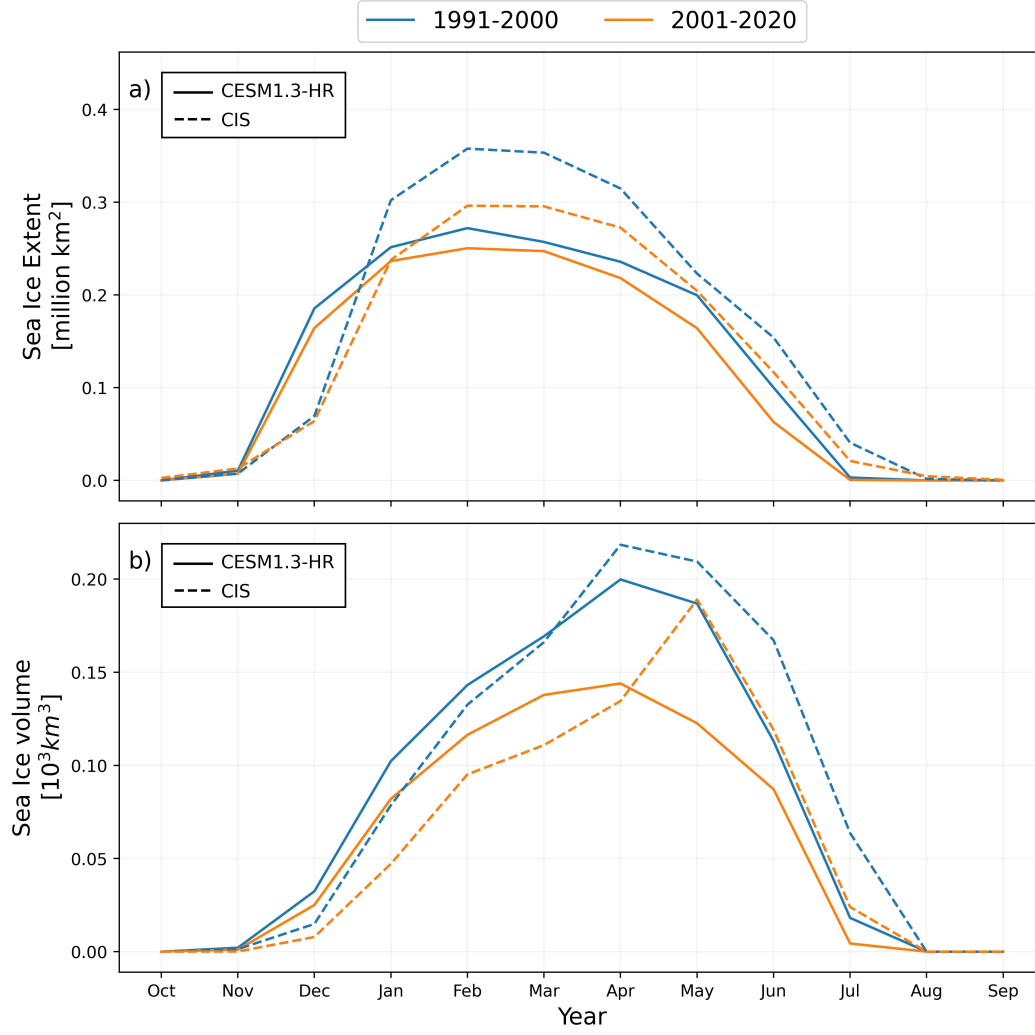


Figure 3. (a) Observed (CIS, dashed) and simulated (CESM1.3-HR, solid) monthly mean sea ice extent (SIE) and (b) volume (SIV) for the 1991-2000 (blue) and 2001-2020 (orange) time period.

timates) with a negative bias in winter, a positive bias in the spring and a $0.05 \times 10^3 km^3$ reduction in April from pre- to post-2000 (see Figure 3b). The simulated inter-annual variability in sea ice volume however is well-captured by the high-resolution CESM when compared with its lower-resolution counterparts.

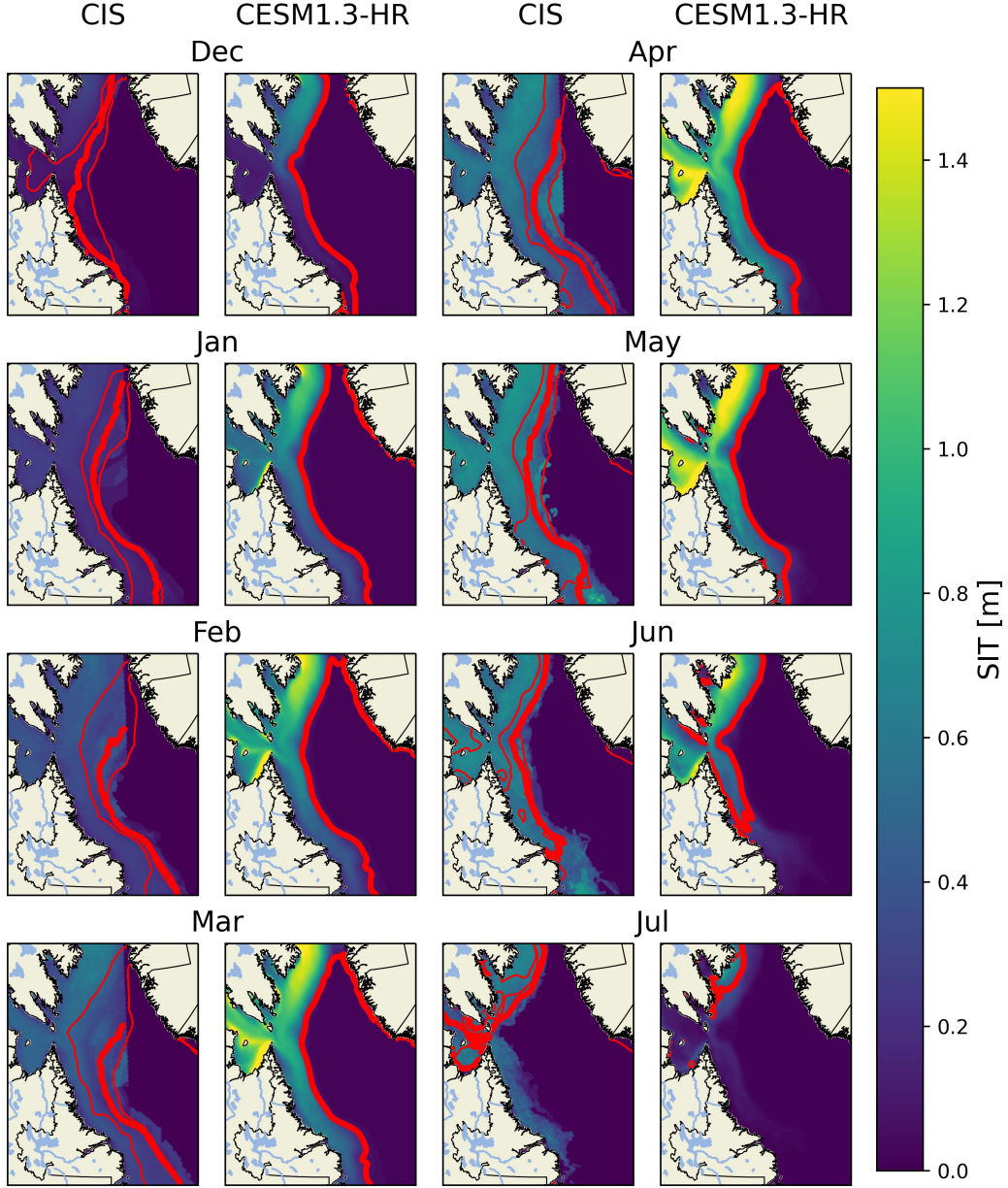


Figure 4. Observed (CIS, 1st and 3rd columns) and simulated (CESM1.3-HR, 2nd and 3rd columns) spatial distribution of monthly climatological SIT for the 1990 - 2020 time period. The bold red contour represents the sea ice edge (SIC=15%), while the thin red contour represents the maximum and minimum SIE from CDR. We use the CDR sea ice extent rather than that of the CIS, as it provides similar values while covering the full Labrador Sea-Baffin Bay region.

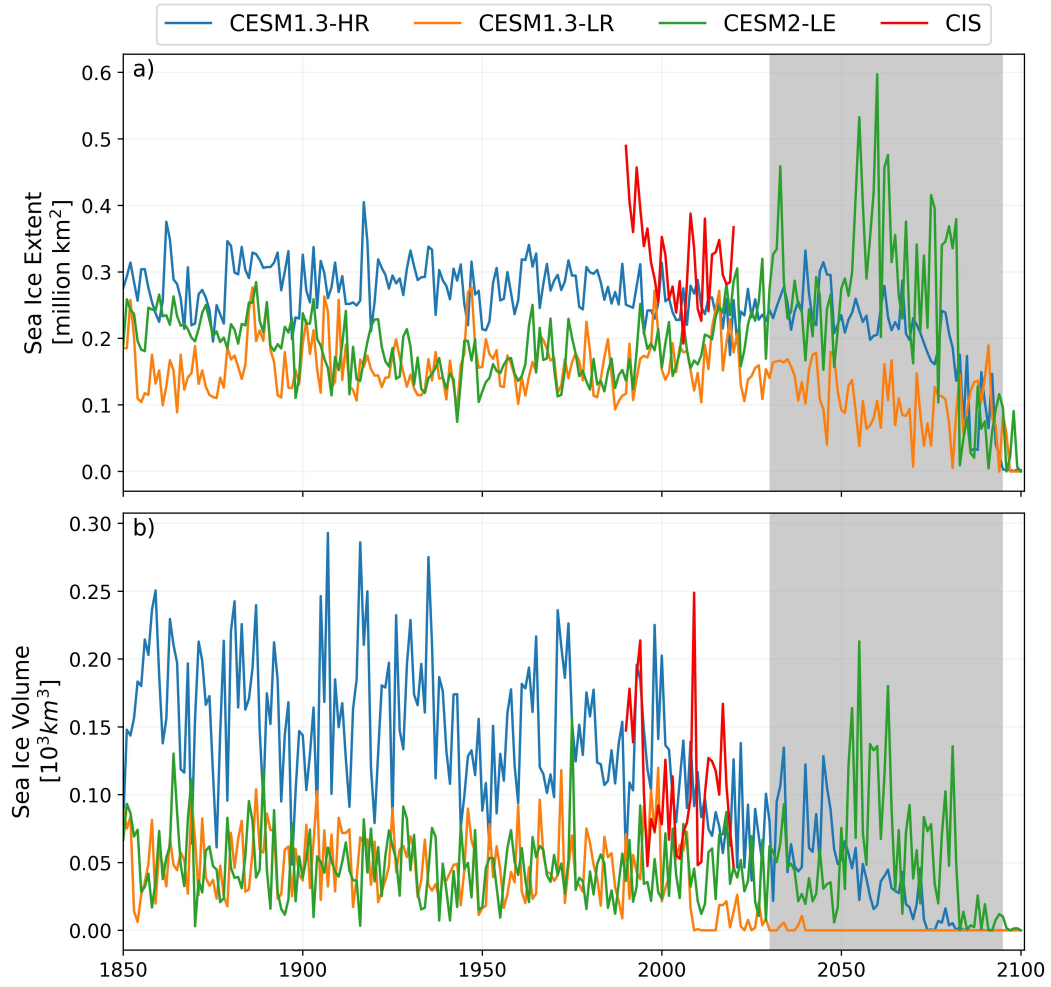


Figure 5. Time series of observed (CIS, red) and simulated (a) March SIE, and (b) May SIV from 1850 to 2100.

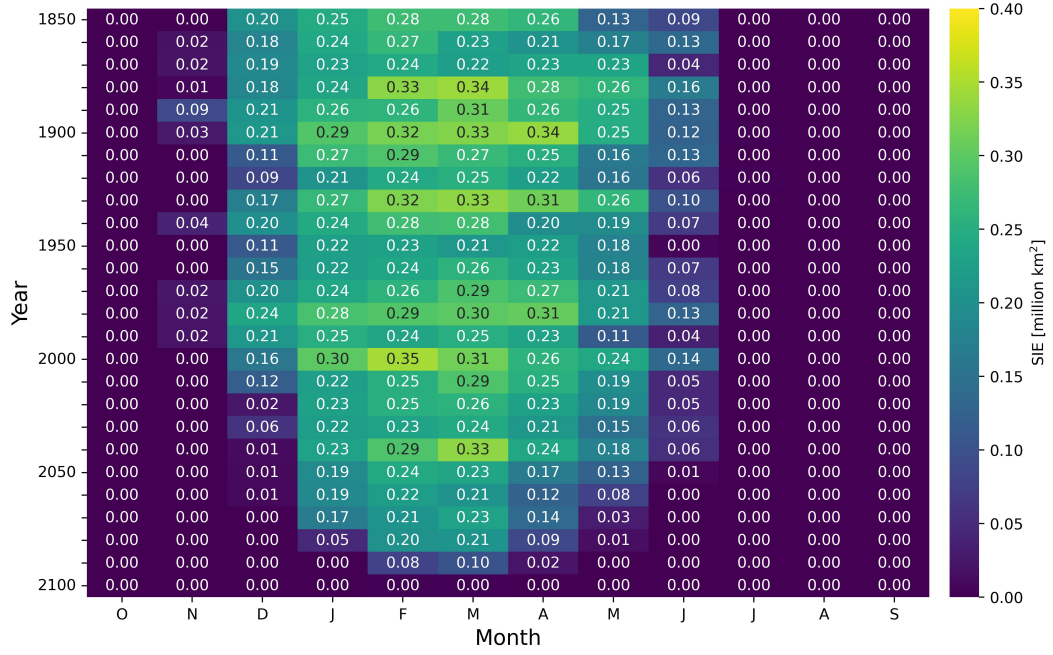


Figure 6. Heat map of simulated (CESM1.3-HR) monthly mean SIE for the period of 1850 to 2100.

4.2 Projections

4.2.1 *Sea Ice Extent (SIE) and Sea Ice Volume (SIV)*

The CESM1.3-HR simulated mean maximum SIE and inter-annual variability in maximum SIV are in general agreement with observations; the simulated variability in maximum SIE however is significantly smaller (50%) in magnitudes (Figure 5). Projections from CESM1.3-HR show a gradual decline in maximum (March) sea ice extent beginning in the mid 20th century, followed by a sharp drop after 2080. When viewed together, these changes give the appearance of a relative stability throughout much of the 20th and 21st centuries, until the rapid decline late in the 21st century (Figure 5, 6). The stability in the simulated maximum SIE is due to southward sea-ice advection from northern Baffin Bay and the Lincoln Sea; conversely, the rapid decline at the end of the 21st century occurs when this source of thicker multi-year sea ice disappears, eliminating advection of sea ice southward through the Nares Strait and along the Labrador Coastline. Note that climate models, including CESM1.3-HR, do not simulate the formation of sea ice arches in the Nares Strait, which in observations often block the southward flow of ice from the Lincoln Sea into Baffin Bay. These arches, once regularly observed, have become increasingly rare in recent years as Arctic sea ice has thinned, suggesting that models may already resemble a future state in which such arches can no longer form. The simulated decline in sea ice volume is more gradual, as is generally observed in models of ice-covered seas, where thinning preconditions the system for rapid future declines (Auclair & Tremblay, 2018).

In the coarser resolution CESM1.3-LR and CESM2-LE, the mean and inter-annual variability in sea ice extent and volume are significantly under-estimated (50%), with a much reduced ice cover in the southern part of the domain (results not shown). Of particular interest is the increase in the mid-21st century, reaching a maximum near 2080, followed by a free-fall to ice-free conditions at the end of the century. This difference in behavior is related primarily to the spatial resolution of the models. In CESM1.3-HR, the Labrador Coastal current, which transports fresher water from sea ice melt, and to a lesser extent from the Greenland ice sheet, remains confined in the coastal current. In contrast, in the coarser-resolution models (CESM2-LE and CESM1.3-LR), this fresher water mixes with offshore waters, resulting in an increase in stratification, an absence of deep water formation and more favorable conditions for sea ice formation (Figure 5,7).

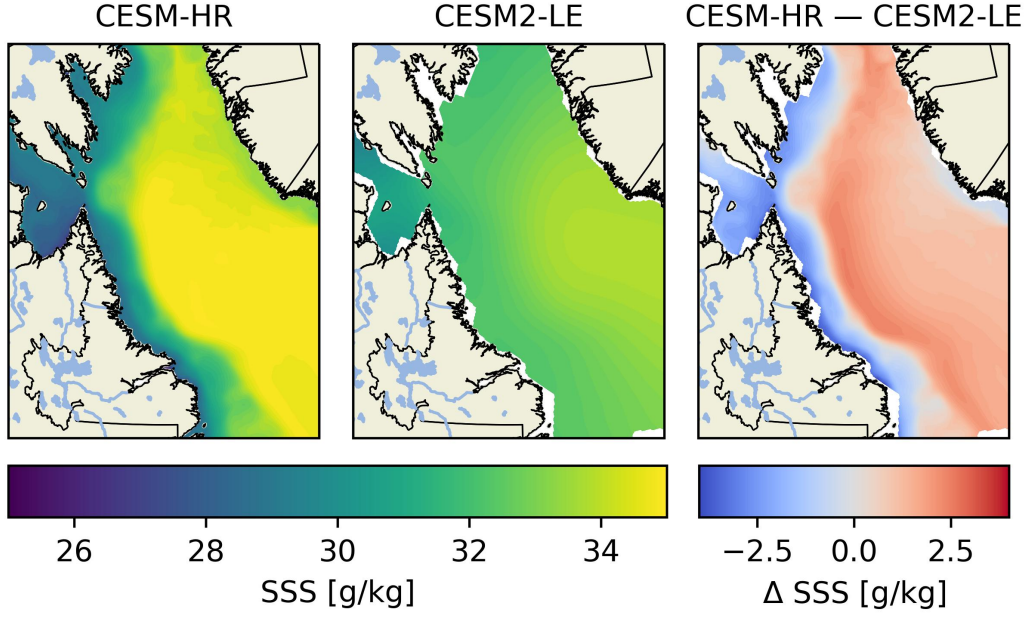


Figure 7. Spatial distribution of March sea surface salinity (SSS) for the period of 2060 to 2070 from high- and low resolution models (CESM1.3-HR, left and CESM2-LE, middle, respectively), and the difference between CESM1.3-HR and CESM2-LE (right).

This change in behaviour as spatial resolution increases was also noted in regional coupled ice-ocean model experiments (e.g, Gou et al., 2022; Dukhovskoy et al., 2019). This shows the importance of resolving fine-scale processes in climate models for regional climate projections.

4.2.2 Length of ice season in coastal communities

The duration of the usable ice season for Labrador’s coastal communities is estimated using a simple Degree Day model—which tracks cumulative temperature deviations from freezing—combined with a one-dimensional, steady-state thermodynamic sea ice model. The season’s start is defined as the date when sea ice reaches a thickness of 30 cm, and the end is marked by the onset of surface melt. For the ice onset date, the first $\sim 240^\circ\text{C} \cdot \text{day}$ ($\pm 80^\circ\text{C} \cdot \text{day}$) are ignored to account for the number of FDD required to bring the surface water temperature to freezing point and for a sea ice cover to first appear as per the Canadian Ice Service ice charts (Figure 8). This delay in freeze-onset is assumed constant in the future for simplicity. The FDD model, forced with surface air temperature from ERA5 as forcing, is first validated against observed sea ice thick-

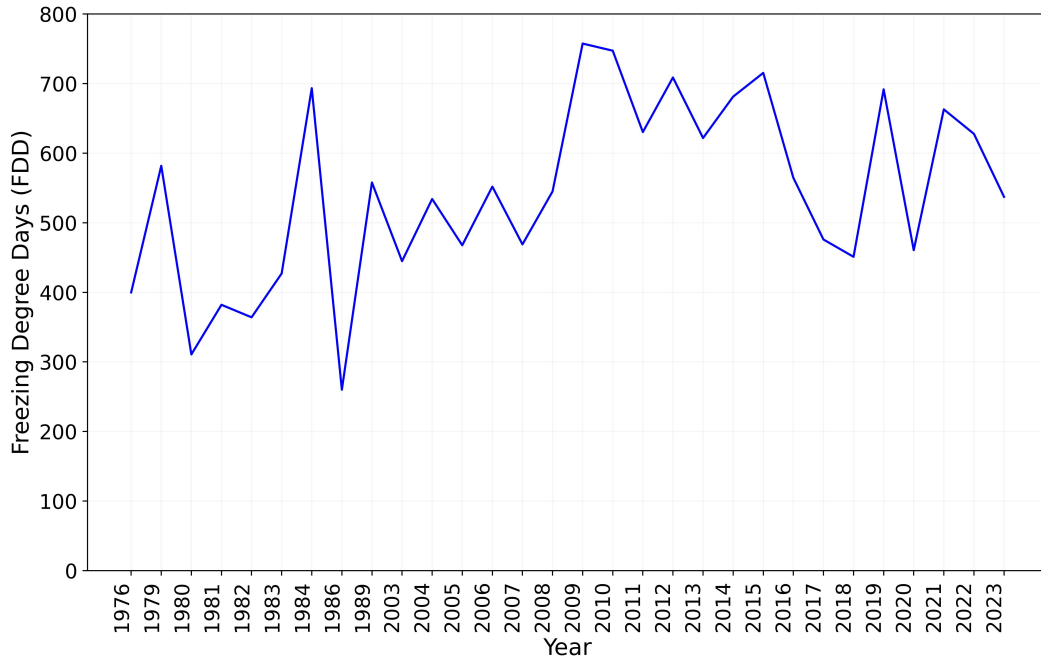


Figure 8. A linear plot of Freezing Degree Days (FDD) as a function of the observed landfast ice appearance date at Nain for all available years from 1976 through 2024.

ness collected in the Nain community, with errors of ± 10 cm, equivalent to 1 day for a surface temperature of -15°C (Figure 9).

The length of the ice season in all five Nunatsiavut coastal communities (Nain, Hopedale, Makkovik, Postville, and Rigolet) start to decline in the early 20th century from ~ 150 days to ~ 50 -70 days at the end of the 21st century, with a relatively small latitudinal dependence (Figure 10). This signal is robust between high and low resolution climate models. In all five locations, interannual variability in the length of the ice season increases in the 21st century by ~ 5 to ~ 15 days with a large increase in decadal variability (figure 10, red curves).

5 Conclusion and Future Work

Sea ice is the main travel route for the people of Nunatsiavut, serving not only as a practical means of transportation but also as a vital link to their cultural and spiritual connections with the land and sea. In this study, we present projections of ice season length for coastal communities of Nunatsiavut from both low- and high-resolution

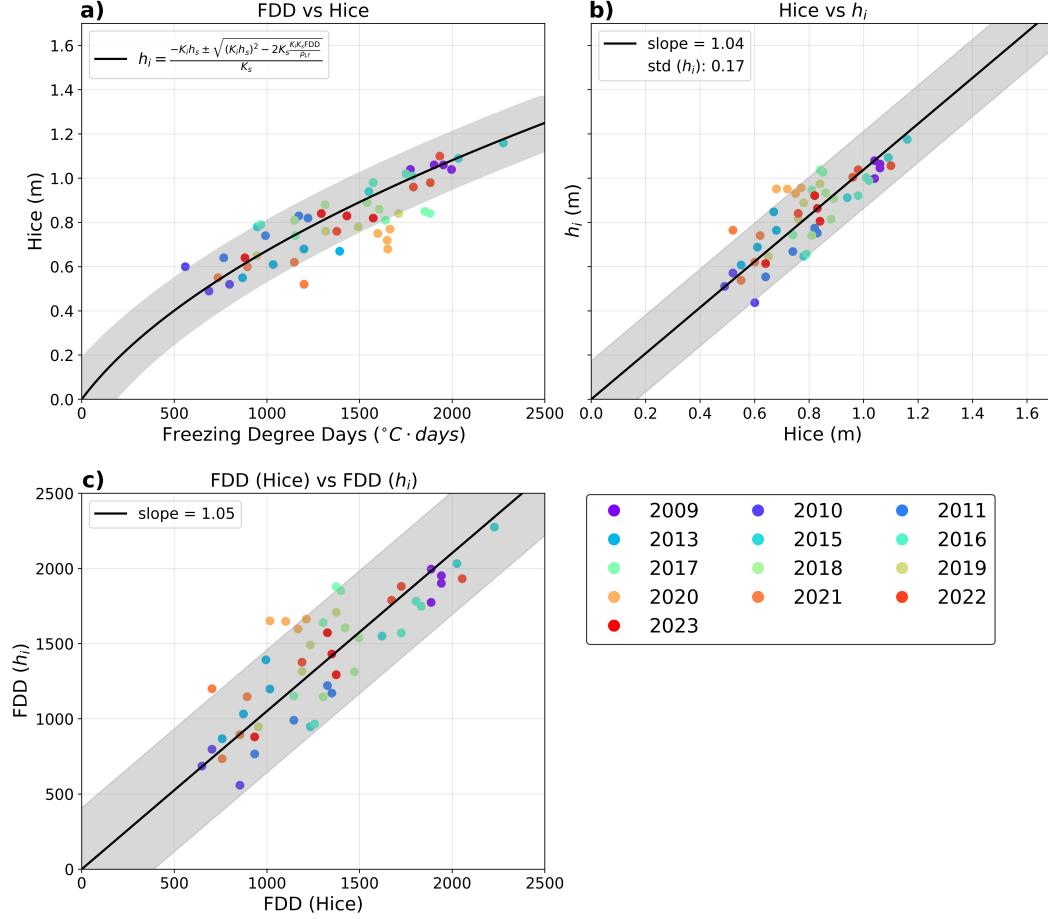


Figure 9. (a) In-situ sea ice thickness (Hice) measurements made by the Nunatsiavut Research Center as a function of Freezing Degree Days (FDD) for the period of 2009-2023. The black line and gray shading show the analytical solution (Equation 6) derived from FDD and the error bar considering the uncertainty of 7 days on the onset of sea ice formation. (b) Scatter plot of observed (Hice) and FDD-derived (h_i) ice thickness. The best line fit forced to pass through the origin has a slope of 1.04 with a standard deviation of 15 cm. (c) Scatter plot of observed FDD (Hice) and FDD-derived (h_i) ice thickness. The best line fit forced to pass through the origin has a slope of 1.05.

Community Earth System Models, namely the CESM1.3-LR, CESM2-LE, and CESM1.3-HR.

Results show that the maximum March sea ice extent remains remarkably stable until approximately 2060, after which a rapid transition to ice-free conditions occurs by the end of the century. This stability is attributed to the continued southward advection of sea ice from the northern Baffin Bay and Lincoln Sea. In contrast, projections from the coarser resolution CESM2-LE model show an increase in sea ice extent starting in the mid-21st century, followed by a sudden collapse. The difference in behaviour

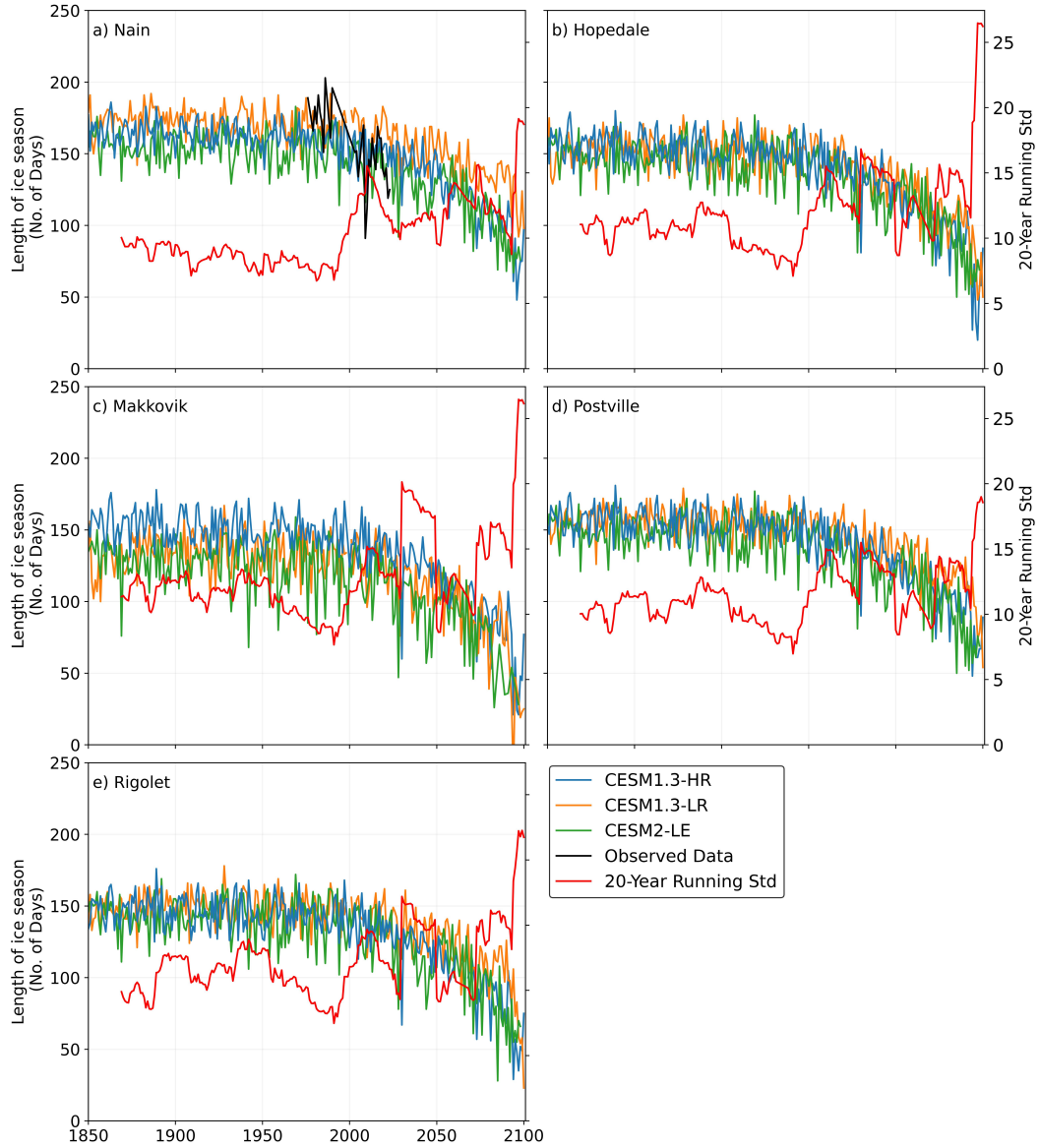


Figure 10. Time series of the length of the ice season (number of days) and 20-year running standard deviation (red) for five locations in Labrador: (a) Nain, (b) Hopedale, (c) Mikkovik, (d) Postville, and (e) Rigolet. The data compares simulations from three different models: CESM1.3-HR (blue), CESM1.3-LR (orange), and CESM2-LE (green), covering the period from 1850 to 2100. Observed data for Nain (black) is included to compare model simulations with historical records for the period 1976–2024.

is linked with the fate of freshwater from sea ice melt in the coastal Labrador Current that mixes laterally in the low-resolution model, and shuts down convection, leading to a re-expansion of the maximum sea ice extent. In the high-resolution model, instead, freshwater remains coastally trapped, and deep convection remains active during the transition to an ice-free Arctic. This difference in behavior highlights the importance of high-

resolution modeling in capturing fine-scale processes essential for accurate regional climate projections.

Given that landfast ice and fjords along the Labrador Coastlines are not well resolved in the low- and high-resolution models, changes in the length of the ice season is estimated from a simple ice-thickness – freezing degree day (FDD) relationship derived from in-situ ice thickness measurements from Nain and Canadian Ice charts for onset of the ice season. Results also show a gradual decline in sea ice thickness and the length of the ice season starting early in the 21st century with sporadic large ice increase in inter-annual variability. This decline begins from an initial mean thickness of 1 meter and a season length of 175 days, gradually decreasing to 0.5 meters and 75 days before the sharp decline in maximum sea ice extent occurs. This reduction is associated with later freeze-up (60%) and earlier breakup (40%) of sea ice, driven by increasing air and sea temperatures.

Landfast ice parameterizations are included in the Community Ice code (CICE). Using these capabilities could allow for a direct assessment of changes in the length of the ice season and a comparison of the simple FDD and simulated ice thickness fields using consistent simulated data sets from the Community Earth System model.

6 Data Availability Statement

The model output (CESM1.3-HR) used in this work is publicly available from the iHESP data archive portal (https://ihesp.github.io/archive/products/ds.archive/Sunway_Runs.html) and documented in Chang et al. (2020). The CESM2-LE data is available on the NSF National Center for Atmospheric Research platform (<https://www.earthsystemgrid.org/dataset/ucar.cgd.cesm2le.output.html>). The CIS gridded version of ice charts is not public but can be made available to the community upon request. The observed Nain data can also be made publicly available to the community upon request. The NSIDC-CDR SIC (https://noaadata.apps.nsidc.org/NOAA/G02202_V4/north/) and ERA5 reanalysis data (<https://www.ecmwf.int/en/forecasts/datasets/reanalysis-datasets/era5>) are available online. All scripts used to obtain results are publicly available at <https://github.com/devThom-studios/Sea-Ice-Projection-Nunatsiavut>.

Acknowledgments

Kyeimiah was supported by a Grant and Contribution from the Canadian Ice Service awarded to Tremblay. This project is also a contribution to a Natural Science and Engineering Research Council (NSERC) Discovery Grant and a NASA ROSE grant (Grant 80NSSC20K1259), both awarded to Tremblay. The CESM1.3-HR and CESM1.3-LR data was kindly provided by Jaison Kurian from the Department of Oceanography of the Texas A&M University.

References

- ACIA. (2004). *Impacts of a Warming Climate: Arctic Climate Impact Assessment*. Cambridge: Cambridge University Press.
- Agnew, T., & Howell, S. (2003). The use of operational ice charts for evaluating passive microwave ice concentration data. *Atmosphere-ocean*, 41(4), 317–331.
- Auclair, G., & Tremblay, L. B. (2018). The role of ocean heat transport in rapid sea ice declines in the community earth system model large ensemble. *Journal of Geophysical Research: Oceans*, 123(12), 8941–8957.
- Bailey, D., Holland, M., Hunke, E., Lipscomb, B., Briegleb, B., Bitz, C., & Schramm, J. (2011). *Community Ice Code (CICE) user's guide version 4.0 released with CCSM 4.0* (Tech. Rep.). Los Alamos, New Mexico: Los Alamos National Laboratory.
- Bitz, Holland, Weaver, A., & Eby, M. (2001, 02). Simulating the ice-thickness distribution in a coupled climate model. *Journal of Geophysical Research*, 106, 2441-2463. doi: 10.1029/1999JC000113
- Bitz, & Lipscomb, W. H. (1999). An energy-conserving thermodynamic model of sea ice. *Journal of Geophysical Research: Oceans*, 104(C7), 15669-15677. Retrieved from <https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/1999JC900100> doi: <https://doi.org/10.1029/1999JC900100>
- Cavalieri, D. J., Gloersen, P., & Campbell, W. J. (1984). Determination of sea ice parameters with the Nimbus 7 SMMR. *Journal of Geophysical Research: Atmospheres*, 89(D4), 5355–5369.
- Cavalieri, D. J., Parkinson, C. L., Gloersen, P., & Zwally, H. J. (1997). *Arctic and Antarctic Sea Ice Concentrations from Multichannel Passive-Microwave Satellite Data Sets: October 1978-September 1995 User's Guide* (Tech. Rep.).

- 430 CBC News. (2021, 3 24). *The sea ice in northern Labrador is thinning — fast.*
 431 *Here's why the Inuit are worried.* [https://www.cbc.ca/news/canada/](https://www.cbc.ca/news/canada/newfoundland-labrador/labrador-nunatsiavut-sea-ice-1.5951551)
 432 [newfoundland-labrador/labrador-nunatsiavut-sea-ice-1.5951551](https://www.cbc.ca/news/canada/newfoundland-labrador/labrador-nunatsiavut-sea-ice-1.5951551).
- 433 CESM2-LE. (2023). *Community Earth System Model Project Large Ensemble Phase*
 434 *2*. Retrieved from [https://www.cesm.ucar.edu/community-projects/lens2#](https://www.cesm.ucar.edu/community-projects/lens2#figures)
 435 [figures](https://www.cesm.ucar.edu/community-projects/lens2#figures) (Accessed: July 2023)
- 436 Chang, P., Zhang, S., Danabasoglu, G., Yeager, S. G., Fu, H., Wang, H., ... oth-
 437 ers (2020). An unprecedented set of high-resolution earth system simulations
 438 for understanding multiscale interactions in climate variability and change.
 439 *Journal of Advances in Modeling Earth Systems*, 12(12), e2020MS002298.
- 440 Close, S., Herbaut, C., Houssais, M.-N., & Blaizot, A.-C. (2018). Mechanisms of
 441 interannual-to decadal-scale winter Labrador Sea ice variability. *Climate dy-*
 442 *namics*, 51(7), 2485–2508. Retrieved from <https://doi.org/10.1007/s00382>
 443 [-017-4024-z](https://doi.org/10.1007/s00382)
- 444 Comiso. (2023). *Bootstrap Sea Ice Concentrations from Nimbus-7 SMMR and*
 445 *DMSP SSM/I-SSMIS, Version 4*. NASA National Snow and Ice Data Center
 446 Distributed Active Archive Center. Retrieved from [https://nsidc.org/data/](https://nsidc.org/data/NSIDC-0079/versions/4)
 447 [NSIDC-0079/versions/4](https://nsidc.org/data/NSIDC-0079/versions/4) doi: 10.5067/X5LG68MH013O
- 448 Comiso, J. (1995). *SSM/I Concentrations Using the Bootstrap Algorithm, NASA*
 449 *Reference Publication 1380*. NASA, Goddard Space Flight Center Greenbelt,
 450 Maryland.
- 451 Comiso, J. C. (1986). Characteristics of Arctic winter sea ice from satellite mul-
 452 tispectral microwave observations. *Journal of Geophysical Research: Oceans*,
 453 91(C1), 975–994.
- 454 Comiso, J. C., & Nishio, F. (2008). Trends in the sea ice cover using enhanced
 455 and compatible AMSR-E, SSM/I, and SMMR data. *Journal of Geophysical*
 456 *Research: Oceans*, 113(C2).
- 457 Cuerrier, A., Brunet, N. D., Gérin-Lajoie, J., Downing, A., & Lévesque, E. (2015).
 458 The study of Inuit knowledge of climate change in Nunavik, Quebec: a mixed
 459 methods approach. *Human ecology*, 43, 379–394.
- 460 Cunsolo, A., Shiwak, I., Wood, M., & Team, I.-I. (2017). “You Need to Be a Well-
 461 Rounded Cultural Person”: youth mentorship programs for cultural preser-
 462 vation, promotion, and sustainability in the Nunatsiavut Region of Labrador.

- 463 *Northern sustainabilities: understanding and addressing change in the circum-*
 464 *polar world*, 285–303.
- 465 Danabasoglu, G., Yeager, S. G., Kwon, Y.-O., Tribbia, J. J., Phillips, A. S., & Hur-
 466 rell, J. W. (2012). Variability of the Atlantic meridional overturning circulation
 467 in CCSM4. *Journal of climate*, 25(15), 5153–5172.
- 468 Deser, C., Holland, M., Reverdin, G., & Timlin, M. (2002). Decadal variations in
 469 Labrador Sea ice cover and North Atlantic sea surface temperatures. *Journal*
 470 *of Geophysical Research: Oceans*, 107(C5), 3–1.
- 471 Deser, C., Magnusdottir, G., Saravanan, R., & Phillips, A. (2004). The effects of
 472 North Atlantic SST and sea ice anomalies on the winter circulation in CCM3.
 473 Part II: Direct and indirect components of the response. *Journal of Climate*,
 474 17(5), 877–889.
- 475 Dukhovskoy, D., Yashayaev, I., Proshutinsky, A., Bamber, J., Bashmachnikov, I.,
 476 Chassignet, E., ... Tedstone, A. (2019). Role of Greenland freshwater anomaly
 477 in the recent freshening of the subpolar North Atlantic. *Journal of Geophysical*
 478 *Research: Oceans*, 124(5), 3333–3360.
- 479 ECCC (Environment and Climate Change Canada). (2023). *Sea ice*. Available
 480 at: [https://www.canada.ca/en/environment-climate-change/services/](https://www.canada.ca/en/environment-climate-change/services/environmental-indicators/sea-ice.html)
 481 [environmental-indicators/sea-ice.html](https://www.canada.ca/en/environment-climate-change/services/environmental-indicators/sea-ice.html). (Accessed August 12, 2024)
- 482 Eden, C., & Willebrand, J. (2001). Mechanism of interannual to decadal variability
 483 of the North Atlantic circulation. *Journal of Climate*, 14(10), 2266–2280.
- 484 Eyring, V., Bony, S., Meehl, G. A., Senior, C. A., Stevens, B., Stouffer, R. J., &
 485 Taylor, K. E. (2016). Overview of the Coupled Model Intercomparison Project
 486 Phase 6 (CMIP6) experimental design and organization. *Geoscientific Model*
 487 *Development*, 9(5), 1937–1958.
- 488 Felt, L., Procter, A. H., & Natcher, D. C. (2012). *Settlement, subsistence, and*
 489 *change among the Labrador Inuit: The Nunatsiavummiut experience* (Vol. 2).
 490 Univ. of Manitoba Press.
- 491 Galley, R. J., Else, B. G., Howell, S. E., Lukovich, J. V., & Barber, D. G. (2012).
 492 Landfast sea ice conditions in the Canadian Arctic: 1983–2009. *Arctic*, 133–
 493 144.
- 494 Gearheard, S., Matumeak, W., Angutikjuaq, I., Maslanik, J., Huntington, H. P.,
 495 Leavitt, J., ... Barry, R. G. (2006). “It’s not that simple”: a collaborative

- comparison of sea ice environments, their uses, observed changes, and adaptations in Barrow, Alaska, USA, and Clyde River, Nunavut, Canada. *AMBIO: A Journal of the Human Environment*, 35(4), 203–211.
- Glissenaar, I. A., Landy, J. C., Petty, A. A., Kurtz, N. T., & Stroeve, J. C. (2021). Impacts of snow data and processing methods on the interpretation of long-term changes in Baffin Bay early spring sea ice thickness. *The Cryosphere*, 15(10), 4909–4927.
- Gou, R., Pennelly, C., & Myers, P. G. (2022). The changing behavior of the West Greenland Current system in a very high-resolution model. *Journal of Geophysical Research: Oceans*, 127(8), e2022JC018404.
- Häkkinen, S. (2002). Surface salinity variability in the northern North Atlantic during recent decades. *Journal of Geophysical Research: Oceans*, 107(C12), SRF4.
- Han, G., Ma, Z., Long, Z., Perrie, W., & Chassé, J. (2019). Climate change on Newfoundland and Labrador shelves: Results from a regional downscaled ocean and sea-ice model under an A1B forcing scenario 2011–2069. *Atmosphere-Ocean*, 57(1), 3–17.
- Hersbach, H., Bell, B., Berrisford, P., Biavati, G., Horányi, A., Muñoz Sabater, J., ... Thépaut, J.-N. (2023). *Era5 hourly data on single levels from 1940 to present*. Copernicus Climate Change Service (C3S) Climate Data Store (CDS). Retrieved from <https://doi.org/10.24381/cds.adbb2d47> (Accessed on 26-Jun-2026)
- Heyblom, K. B., Singh, H. A., Rasch, P. J., & DeRepentigny, P. (2022). Increased variability of biomass burning emissions in CMIP6 amplifies hydrologic cycle in the CESM2 Large Ensemble. *Geophysical Research Letters*, 49(5), e2021GL096868.
- Howell, S., Babb, D., Landy, J. C., Moore, G., Montpetit, B., & Brady, M. (2023). A comparison of Arctic Ocean sea ice export between Nares Strait and the Canadian Arctic Archipelago. *Journal of Geophysical Research: Oceans*, 128(4), e2023JC019687.
- Hunke, E., & Dukowicz, J. (2002, 07). The Elastic Viscous Plastic Sea Ice Dynamics Model in General Orthogonal Curvilinear Coordinates on a Sphere—Incorporation of Metric Terms. *Monthly Weather Review - MON WEATHER*

- 529 *REV*, 130. doi: 10.1175/1520-0493(2002)130(1848:TEVPSI)2.0.CO;2
- 530 Hunke, E., Lipscomb, W. H., Turner, A. K., Jeffery, N., & Elliott, S. (2010). Cice:
531 the los alamos sea ice model documentation and software user's manual version
532 4.1 la-cc-06-012. *T-3 Fluid Dynamics Group, Los Alamos National Laboratory*,
533 675, 500.
- 534 JCOMM. (2004). *SIGRID 3: A Vector Archive Format for Sea Ice Charts* (Tech.
535 Rep. No. 23). Geneva, Switzerland: Joint WMO-IOC Technical Commission
536 for Oceanography and Marine Meteorology.
- 537 Kern, S., Laverne, T., Notz, D., Pedersen, L. T., & Tonboe, R. (2020). Satellite
538 passive microwave sea-ice concentration data set inter-comparison for Arctic
539 summer conditions. *The Cryosphere*, 14(7), 2469–2493.
- 540 Kwok, R. (2005). Variability of Nares Strait ice flux. *Geophysical Research Letters*,
541 32(24).
- 542 Large, W. G., McWilliams, J. C., & Doney, S. C. (1994). Oceanic vertical
543 mixing: A review and a model with a nonlocal boundary layer param-
544 eterization. *Reviews of Geophysics*, 32(4), 363–403. Retrieved from
545 <https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/94RG01872>
546 doi: <https://doi.org/10.1029/94RG01872>
- 547 Lawrence, D. M., Oleson, K. W., Flanner, M. G., Thornton, P. E., Swenson, S. C.,
548 Lawrence, P. J., ... others (2011). Parameterization improvements and func-
549 tional and structural advances in version 4 of the Community Land Model.
550 *Journal of Advances in Modeling Earth Systems*, 3(1).
- 551 Lipscomb, W. H. (2001). Remapping the thickness distribution in sea ice models.
552 *Journal of Geophysical Research: Oceans*, 106(C7), 13989–14000. Retrieved
553 from [https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/](https://agupubs.onlinelibrary.wiley.com/doi/abs/10.1029/2000JC000518)
554 [2000JC000518](https://doi.org/10.1029/2000JC000518) doi: <https://doi.org/10.1029/2000JC000518>
- 555 Mahoney, A., Gearheard, S., Oshima, T., & Qillaq, T. (2009). Sea ice thickness mea-
556 surements from a community-based observing network. *Bulletin of the Ameri-
557 can Meteorological Society*, 90(3), 370–378.
- 558 Meehl, G. A., Yang, D., Arblaster, J. M., Bates, S. C., Rosenbloom, N., Neale, R.,
559 ... others (2019). Effects of model resolution, physics, and coupling on South-
560 ern Hemisphere storm tracks in CESM1. 3. *Geophysical Research Letters*,
561 46(21), 12408–12416.

- Meier, Fetterer, Windnagel, & Stewart. (2021). *NOAA/NSIDC Climate Data Record of Passive Microwave Sea Ice Concentration, Version 4*. National Snow and Ice Data Center. Retrieved from <https://nsidc.org/data/G02202/versions/4> doi: 10.7265/efmz-2t65
- Münchow, A. (2016). Volume and freshwater flux observations from Nares Strait to the west of Greenland at daily time scales from 2003 to 2009. *Journal of Physical Oceanography*, 46(1), 141–157.
- Neale, R. B., Chen, C.-C., Gettelman, A., Lauritzen, P. H., Park, S., Williamson, D. L., ... others (2010). Description of the NCAR community atmosphere model (CAM 5.0). *NCAR Tech. Note NCAR/TN-486+ STR*, 1(1), 1–12.
- Newswire. (2021, 8 16). *Canada invests in Inuit-led climate adaptation and conservation in Nunatsiavut, Newfoundland and Labrador*. <https://www.newswire.ca/news-releases/canada-invests-in-inuit-led-climate-adaptation-and-conservation-in-nunatsiavut-newfoundland-and-labrador-809891914.html>.
- Nickels, S., Furgal, C., Buell, M., Moquin, H., et al. (2005). *Unikkaaqatigiit: Putting the human face on climate change-Perspectives from Inuit in Canada*. Joint publication of Inuit Tapiriit Kanatami, Nasivvik Centre for Inuit ...
- Notz, D., & Community, S. (2020). Arctic sea ice in CMIP6. *Geophysical Research Letters*, 47(10), e2019GL086749.
- Nunatsiavut Research Center. (2024). *Ice thickness data from nain, nunatsiavut (2009-2023)*. Unpublished raw data. Retrieved from <https://nunatsiavutresearchcentre.com/> (Accessed February 2024)
- Olauson, J. (2018). ERA5: The new champion of wind power modelling? *Renewable energy*, 126, 322–331.
- Ouellet-Bernier, M.-M., de Vernal, A., & Sea, L. (2021). Winter freeze-up and summer break-up in Nunatsiavut, Canada, from 1770 to 1910. *CONTRIBUTION DES SOURCES HISTORIQUES À L'ÉTUDE DU CLIMAT DE LA CÔTE DU LABRADOR/NUNATSIAVUT, 1750 À 1950: UN REGARD SUR LES SOURCES DISCURSIVES, DOCUMENTAIRES ET INSTRUMENTALES*, 221.
- Ouellet-Bernier, M.-M., de Vernal, A., Chartier, D., & Boucher, É. (2021). Historical perspectives on exceptional climatic years at the Labrador/Nunatsiavut coast

- 1780 to 1950. *Quaternary Research*, *101*, 114–128.
- Pearce, T. D., Ford, J. D., Laidler, G. J., Smit, B., Duerden, F., Allarut, M., ... others (2009). Community collaboration and climate change research in the Canadian Arctic. *Polar Research*, *28*(1), 10–27.
- Pizzolato, L., Howell, S. E., Derksen, C., Dawson, J., & Copland, L. (2014). Changing sea ice conditions and marine transportation activity in Canadian Arctic waters between 1990 and 2012. *Climatic change*, *123*, 161–173.
- Rodgers, K. B., Lee, S.-S., Rosenbloom, N., Timmermann, A., Danabasoglu, G., Deser, C., ... others (2021). Ubiquity of human-induced changes in climate variability. *Earth System Dynamics*, *12*(4), 1393–1411.
- Small, R. J., Bacmeister, J., Bailey, D., Baker, A., Bishop, S., Bryan, F., ... others (2014). A new synoptic scale resolving global climate simulation using the Community Earth System Model. *Journal of Advances in Modeling Earth Systems*, *6*(4), 1065–1094.
- Smith, R., Jones, P., Briegleb, B., Bryan, F., Danabasoglu, G., Dennis, J., ... others (2010). The parallel ocean program (POP) reference manual ocean component of the community climate system model (CCSM) and community earth system model (CESM). *LAUR-01853*, *141*, 1–140.
- Tivy, A., Howell, S. E., Alt, B., McCourt, S., Chagnon, R., Crocker, G., ... Yackel, J. J. (2011). Trends and variability in summer sea ice cover in the Canadian Arctic based on the Canadian Ice Service Digital Archive, 1960–2008 and 1968–2008. *Journal of Geophysical Research: Oceans*, *116*(C3).
- Urban, A., Di Napoli, C., Cloke, H. L., Kyselý, J., Pappenberger, F., Sera, F., ... others (2021). Evaluation of the ERA5 reanalysis-based Universal Thermal Climate Index on mortality data in Europe. *Environmental research*, *198*, 111227.
- Vaideanu, P., Stepanek, C., Dima, M., Schrepfer, J., Matos, F., Ionita, M., & Lohmann, G. (2023). Large-scale sea ice–Surface temperature variability linked to Atlantic meridional overturning circulation. *Plos one*, *18*(8), e0290437.
- Van Marle, M. J., Kloster, S., Magi, B. I., Marlon, J. R., Daniau, A.-L., Field, R. D., ... others (2017). Historic global biomass burning emissions for CMIP6 (BB4CMIP) based on merging satellite observations with proxies and fire models (1750–2015). *Geoscientific Model Development*, *10*(9), 3329–3357.

- Wilbert Weijer and Thomas W.N. Haine and Ali H. Siddiqui and Wei Cheng and
 Milena Veneziani and Prajvala Kurtakoti. (2021). Interactions between
 the arctic mediterranean and the atlantic meridional overturning circula-
 tion: A review. *Oceanography*. Retrieved from <https://doi.org/10.1007/s00382-017-4024-z> (Received 23 June 2017, Accepted 21 November 2017, Published 02 December 2017, Issue Date October 2018)
- Wu, Y., Zhao, X., Qi, Z., Zhou, K., & Qiao, D. (2023). Relative Contribution of
 Atmospheric Forcing, Oceanic Preconditioning and Sea Ice to Deep Convection
 in the Labrador Sea. *Journal of Marine Science and Engineering*, 11(4), 869.
- Yashayaev, I. (2024). Intensification and shutdown of deep convection in the
 Labrador Sea were caused by changes in atmospheric and freshwater dy-
 namics. *Communications Earth & Environment*, 5, 156. Retrieved from
<https://doi.org/10.1038/s43247-024-01296-9> (Accessed: 2024-06-29)
 doi: 10.1038/s43247-024-01296-9
- Yashayaev, I., van Aken, H. M., Holliday, N. P., & Bersch, M. (2007). Transfor-
 mation of the Labrador Sea water in the subpolar North Atlantic. *Geophysical
 Research Letters*, 34(22).
- Zhang, S., Fu, H., Wu, L., Li, Y., Wang, H., Zeng, Y., ... Guo, Y. (2020). Optimiz-
 ing high-resolution Community Earth System Model on a heterogeneous many-
 core supercomputing platform. *Geoscientific Model Development*, 13(10),
 4809–4829. Retrieved from <https://gmd.copernicus.org/articles/13/4809/2020/> doi: 10.5194/gmd-13-4809-2020
- Zhu, J., Xie, A., Qin, X., Wang, Y., Xu, B., & Wang, Y. (2021). An assessment
 of ERA5 reanalysis for Antarctic near-surface air temperature. *Atmosphere*,
 12(2), 217.